

Your grade: 100%

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Next item →

1. **True or False:** The defining feature of Bayesian inference is that it allows for probability distributions over statistical parameters (as well as over data). 2 / 2 points

- ☒ True
- ☐ False
- ☒ Nice work

2. **True or False:** In Bayesian methods, statistical parameters are unknown and fixed (i.e., not random). 2 / 2 points

- ☐ True
- ☒ False
- ☒ Nice work

3. **True or False:** Frequentist hypothesis testing results in a new probability distribution over the parameter. It represents our belief in the value of the parameter after accounting for evidence (the sample). 2 / 2 points

- ☐ True
- ☒ False
- ☒ Nice work

4. **True or False:** The maximum a posteriori estimator provides a method for estimating the entire posterior distribution of a Bayesian inference problem. 2 / 2 points

- ☐ True
- ☒ False
- ☒ Nice work

5. **True or False:** In frequentist methods, statistical parameters are unknown and fixed (i.e., not random). 2 / 2 points

- ☐ True
- ☐ False
- ☒ Nice work

6. **True or False:** Bayes' theorem is a mathematically rigorous way to combine our prior belief about a parameter with data from a sample. 2 / 2 points

- ☒ True
- ☐ False
- ☒ Nice work

7. In Bayesian statistics, what does the posterior distribution represent? 2 / 2 points

- ☒ The updated beliefs about parameter values after observing data.
- ☐ The distribution of the sample data itself.
- ☐ The likelihood of the observed data given certain parameter values.
- ☐ The prior beliefs about parameter values before observing data.

☒ Nice work
Correct! The posterior distribution combines the prior distribution and likelihood to reflect updated beliefs.

8. **True or False:** The maximum a posteriori estimator provides a method for estimating the expected value of the posterior distribution of a Bayesian inference problem. 2 / 2 points

- ☐ True
- ☒ False
- ☒ Nice work

9. Ethel is collecting data on the sale of bicycles for a national retailer. The purpose of her analysis is to infer how bicycle sales vary by geographic location (e.g., do more expensive bikes sell better in certain locations?). Imagine a dataset of $n = 100$ receipts, $r = 15$ of which contain the sale of a bike over \$1,000. We are unsure of the stopping rule that Ethel used to obtain this dataset. 2 / 2 points

Which of the following stopping rules would result in the same output (posterior/estimate) in Bayesian inference?

- Ethel stopped because she observed $n = 100$ sale receipts.
- Ethel stopped because she observes $r = 15$ sales of a bike over \$1,000
- Ethel stopped because she is done with work for the day.
- Ethel stopped because she analyzed all of the receipts from this retailer.

- ☒ 1, 2, 3, 4
- ☐ 1, 2
- ☐ 1, 3, 4
- ☐ 1

☒ Nice work

10. **Select all that apply:** The likelihood function in Bayesian inference ... 2 / 2 points

☒ is a function of the parameter(s) given fixed data.

☒ Nice work

☒ is an input to Bayes' theorem.

☒ Nice work

☐ is a function of the data given fixed parameter(s).

☐ integrates to 1.

11. What is a stopping rule in the context of research data collection and analysis? 2 / 2 points

- ☒ A rule for determining when to stop collecting data during a study.
- ☐ A strategy to pause the study for interim results.
- ☐ A guideline for participants to leave the study at any time.
- ☐ A checkpoint to ensure data integrity before analysis.

☒ Nice work

Correct! A stopping rule provides criteria for when data collection should end, impacting the study's results.

12. Assume that (X,Y) follow a bivariate normal distribution with $E(X) = E(Y) = 0$ and $\text{Var}(X) = \text{Var}(Y) = 1$. The pdf of (X,Y) is given as 2 / 2 points

$$f(x,y|\rho) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right\},$$

where ρ is the correlation coefficient for X and Y . Suppose that we observe **one** realization of (X,Y) . Which of the following represents the likelihood function?

- ☐ None of the above
- ☒ $f(\rho|x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right\}$
- ☐ $f(\rho|x,y) = 1$
- ☐ $f(x,y) = \frac{1}{2\pi} \exp\left\{-\frac{x^2 + y^2}{2}\right\}$

☒ Nice work

Nice!

13. Assume that (X,Y) follow a bivariate normal distribution with $E(X) = E(Y) = 0$ and $\text{Var}(X) = \text{Var}(Y) = 1$. The pdf of (X,Y) is given as 2 / 2 points

$$f(x,y|\rho) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right\},$$

where ρ is the correlation coefficient for X and Y . Suppose that we observe n iid realizations of (X,Y) . This pdf is a bivariate function.

- ☒ True
- ☐ False

☒ Nice work